

SURE TRUST • CAPSTONE PROJECT PRESENTATION

EstateIQ: Gurgaon Real Estate Analytics & Machine Learning Platform

Production-Grade Price Predictor, Content Recommender & Market Intelligence Platform

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Stack: Python • Scikit-Learn • FastAPI • Streamlit • Docker • Render • GitHub Actions

Problem Statement & Real-World Context

1. Market Price Opacity

- **Industry Issue:** Property pricing in Gurgaon is non-transparent and varies heavily across 100+ sectors.
- **High Variance:** Similar properties differ in price by up to 40% due to unstandardised broker estimates.
- **Buyer Risk:** Home buyers risk overpaying without data-driven benchmark valuations.

2. Complex Non-Linear Drivers

- **Multi-Factor Dependency:** Prices depend non-linearly on area, floor height, age, luxury amenities & servant rooms.
- **Location Premium:** Premium corridors (Golf Course Rd, Sec 54) command exponential location surcharges.
- **Interaction Complexity:** Simple linear models fail to capture feature interactions.

3. Limitations of Existing Solutions

- **Broker Subjectivity:** Manual appraisals rely on personal bias and incomplete local knowledge.
- **Listing Averages:** Property portal averages ignore structural differences between builder floors and flats.
- **No Discovery Tool:** Lack of integrated similarity search for finding alternative projects.

4. The EstateIQ Solution

- **Objective Machine Learning:** RandomForest model achieving $R^2 = 0.8211$ on 3,554 cleaned property listings.
- **Content Recommender:** TF-IDF + Cosine Similarity engine for discovery of matching projects.
- **Full-Stack Microservices:** FastAPI REST backend paired with interactive Streamlit web dashboard.

Project Overview & Key Objectives

Core Objectives

- **Predictive Modeling:** Train a robust ML regression model to estimate property values in Crore INR.
- **Apartment Discovery:** Build a text-similarity recommendation engine for project search.
- **Market Intelligence:** Provide locality heatmaps and interactive price analytics.
- **Production Readiness:** Develop a REST API, web UI, Docker assets, and CI/CD pipelines.

Key Features

- **Single Property Valuation:** Instant price estimate based on 12 key property attributes.
- **Batch Prediction:** CSV bulk upload support for portfolio price estimation.
- **TF-IDF Recommender:** Search matching projects by locality, amenities & text query.
- **Interactive Analytics:** Plotly price distribution, sector heatmaps & boxplots.

Target End Users

- **Home Buyers:** Verify fair market prices before submitting purchase offers.
- **Real Estate Investors:** Identify high-value sectors and analyze price-per-sq.ft trends.
- **Property Brokers:** Generate data-backed valuation reports for prospective clients.
- **PropTech Analysts:** Utilize REST API endpoints for downstream property valuation.

Technology Stack & Enterprise Architecture

Frontend Layer

- **Framework:** Streamlit Multi-Page Enterprise App
- **Visualisations:** Plotly Express & Graph Objects
- **Styling:** Custom CSS & Dark Theme Palette

Backend API Layer

- **Web Framework:** FastAPI (Python 3.11/3.14)
- **Data Validation:** Pydantic V2 BaseModel & Aliases
- **ASGI Server:** Uvicorn Async Worker

Machine Learning Engine

- **Pipeline:** Scikit-Learn ColumnTransformer
- **Model:** RandomForestRegressor (500 Trees)
- **Target Scaling:** TransformedTargetRegressor (log1p)

Data & NLP Stack

- **Processing:** Pandas, NumPy, Scipy
- **Text Recommender:** TfidfVectorizer & Cosine Similarity
- **Persistence:** Joblib Model & JSON Metadata

DevOps & Deployment

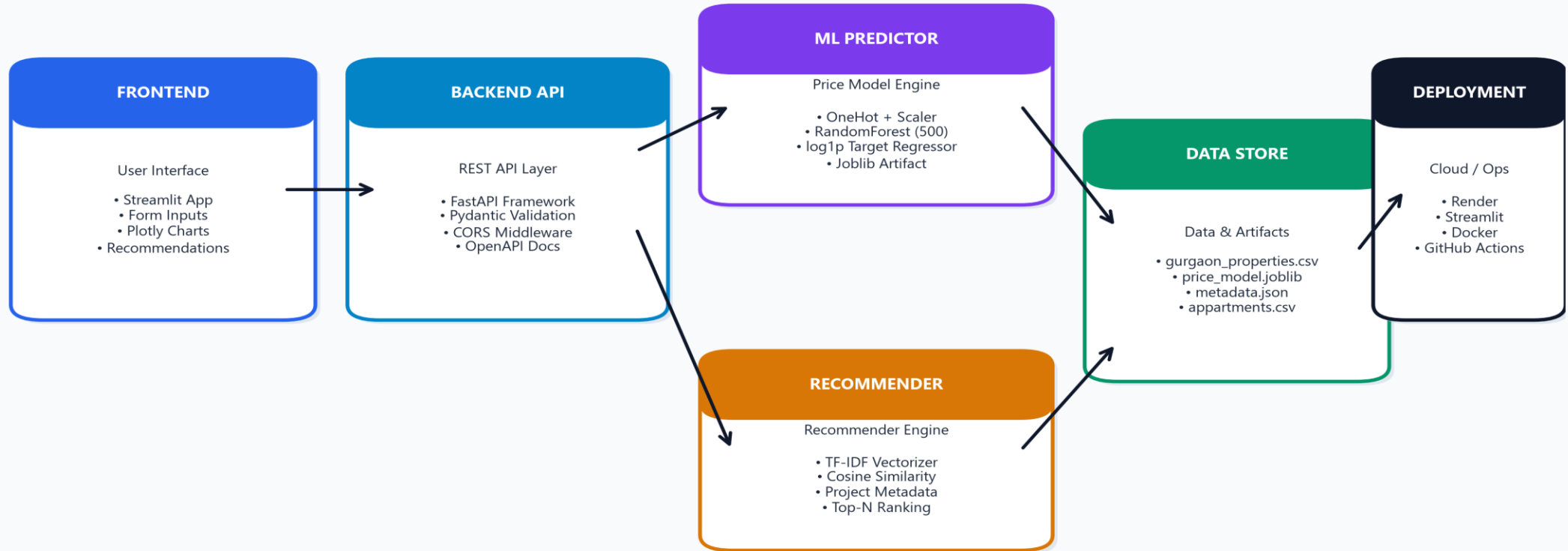
- **Containers:** Docker & Docker Compose
- **Cloud Hosting:** Render (API) & Streamlit Cloud (UI)
- **CI/CD Pipeline:** GitHub Actions Automated Workflows

Quality & Testing

- **Test Suite:** Pytest Unit & API Smoke Tests
- **Code Hygiene:** Ruff Linter & Formatter
- **Type Checking:** Python Type Hints & Annotations

Complete System Architecture

EstateIQ - System Architecture & Microservice Pipeline



Data Collection & Raw Dataset Specs

Raw Data Ingestion & Sources

- **Data Extraction:** Scraped property listings from major Gurgaon real estate portals.
- **houses.csv (2.0 MB):** Independent house listing records containing room & plot details.
- **flats.csv (5.7 MB):** Residential apartment listings with floor and society metadata.
- **apartments.csv (272 KB):** Project-level metadata including facilities, nearby locations & amenities.
- **latlong.csv:** Geospatial sector coordinates for map mapping & analytics.
- **Consolidation:** Merged house & flat datasets into unified residential property table.

Dataset Statistics & Summary

- **Final Modeling Records:** 3,554 Cleaned Property Records
- **Total Features:** 13 Core Features (12 Predictors + 1 Target)
- **Target Variable:** price (Measured in Crore INR)
- **Property Types:** Flats (Apartments) & Independent Houses
- **Geographic Coverage:** 100+ Gurgaon Sectors & Key Corridors
- **Missing Value Standard:** Imputed via sector mode & median area calculations.

Database & Data Schema Design

Properties Entity Schema

- **property_type:** Categorical: 'flat' or 'house'
- **sector:** Categorical: Gurgaon Sector (e.g. 'sector 54')
- **bedRoom:** Numeric: Count of bedrooms
- **bathroom:** Numeric: Count of bathrooms
- **balcony:** Categorical: Balcony count ('1', '2', '3', '3+')
- **built_up_area:** Numeric: Total area in square feet
- **servant room / store room:** Binary: 1 if available, 0 otherwise
- **furnishing_type:** Categorical: Unfurnished, Semi, Furnished
- **luxury_category:** Categorical: Low, Medium, High
- **floor_category:** Categorical: Low Floor, Mid Floor, High Floor

Apartments & Geo Schema

- **Apartments Entity:** Used for TF-IDF Text Recommendation Engine
- **PropertyName:** String: Name of apartment project
- **PropertySubName:** String: Developer or sub-locality name
- **NearbyLocations:** List[String]: Nearby landmarks & hubs
- **LocationAdvantages:** List[String]: Connectivity advantages
- **TopFacilities:** List[String]: Club, Pool, Gym, Security, Lifts
- **Link:** String: External property URL link
- **Geo Entity:** Sector, Latitude, Longitude for map rendering

Data Preprocessing & Data Cleaning

1. Unit Standardisation

- **Price Standardisation:** Converted messy text values (e.g. '1.5 Lacs', '2.4 Cr') to uniform Crore INR scale.
- **Sector Normalisation:** Cleaned sector names, removing noise and standardising format (e.g. 'sector 54').
- **Area Extraction:** Parsed built-up area into standardized numeric square feet.

2. Outlier Elimination and Capping

- **Area-Room Anomaly Filter:** Removed listings where area per bedroom was unrealistically small (<150 sq.ft/BHK).
- **Extreme Price Filter:** Treated extreme price outliers that skew regression loss.
- **Duplicate Removal:** Eliminated identical cross-posted listing records.

3. Missing Value Imputation

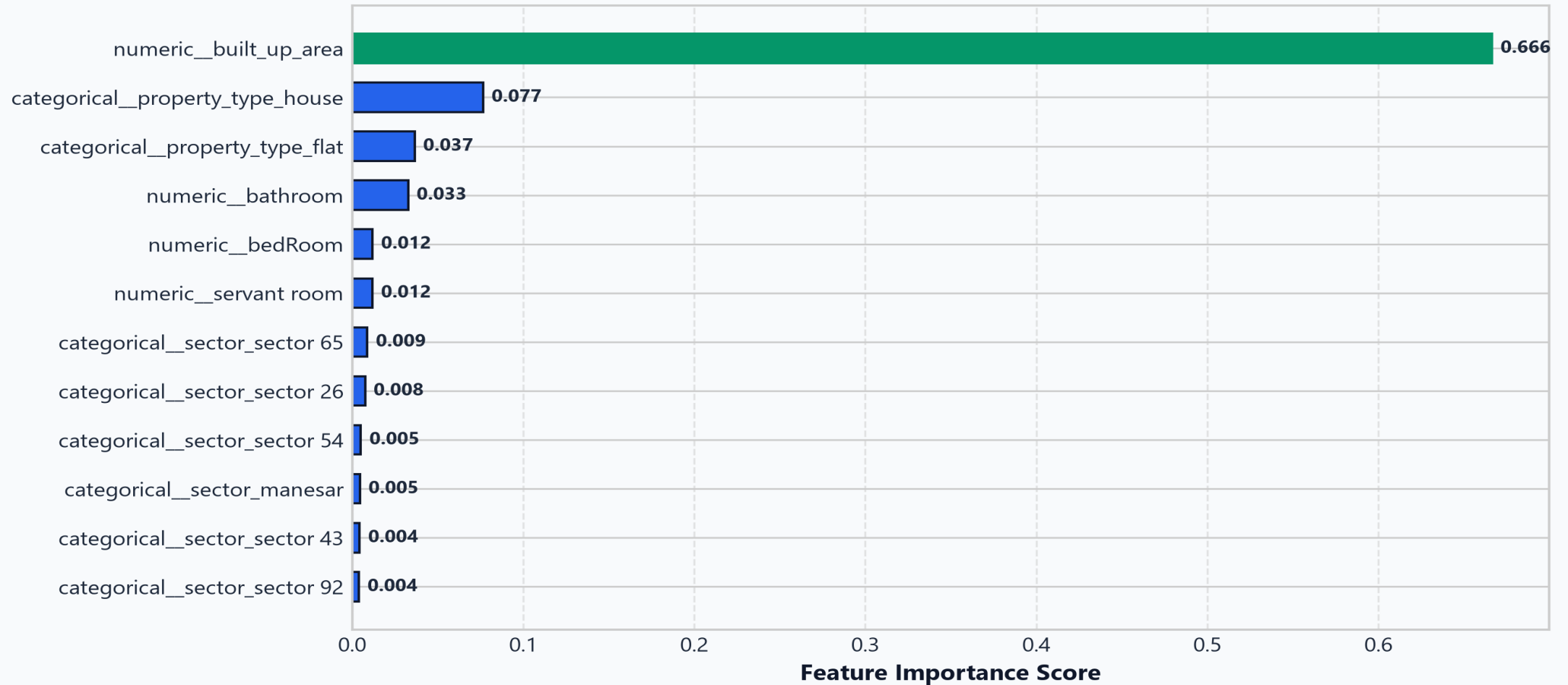
- **Categorical Imputation:** Missing sector and furnishing fields filled with sector-level mode.
- **Numeric Imputation:** Missing area values imputed using median area per BHK and sector.
- **Room Flags:** Missing room indicator text default-filled as 0.

4. Pipeline Integration

- **Sklearn ColumnTransformer:** Applied OneHotEncoder to 7 categorical features and StandardScaler to 5 numerical features.
- **Handle Unknowns:** Set handle_unknown='ignore' on OneHotEncoder for zero-fail API inference.
- **Reproducibility:** Encapsulated preprocessing inside the saved sklearn model pipeline.

Feature Engineering & Feature Importance

Top 12 Features by Random Forest Importance



Train-Test Split Strategy

Split Configuration

- **Training Ratio:** 80% of total dataset (2,843 property records).
- **Testing Ratio:** 20% of total dataset (711 property records).
- **Random Seed:** random_state=42 fixed across all training experiments.
- **Purpose:** Ensures strict separation between model training and performance evaluation.
- **Data Leakage Prevention:** Transformers (Scaler & Encoder) fitted ONLY on X_train.

Why Train-Test Split Matters

- **Generalization Evaluation:** Tests if the model learns general real estate pricing patterns rather than memorizing data.
- **Overfitting Detector:** Compares training vs validation error to detect model over-complexity.
- **Production Simulation:** Simulates real-world API prediction scenarios where incoming property listings are unseen.

Machine Learning Training Pipeline

EstateIQ - Machine Learning Training & Inference Pipeline



Model Selection & Algorithm Comparison

Evaluated ML Algorithms

- **Linear Regression:** Baseline model. Failed to capture non-linear location and area interactions ($R^2 \sim 0.64$).
- **Decision Tree Regressor:** High variance, overfitted training data heavily ($R^2 \sim 0.71$).
- **Random Forest Regressor:** SELECTED MODEL. Ensemble of 500 decision trees. Excellent non-linear handling ($R^2 = 0.8211$).

Why Random Forest Selected

- **Non-Linear Flexibility:** Effortlessly captures complex interaction between sector premiums and built-up area.
- **Robust to Outliers:** Averages predictions across 500 trees, reducing sensitivity to price extremes.
- **Low Tuning Overhead:** Requires less hyperparameter tuning than complex boosting methods.
- **Built-in Feature Importance:** Provides explicit feature importance scores for model explainability.

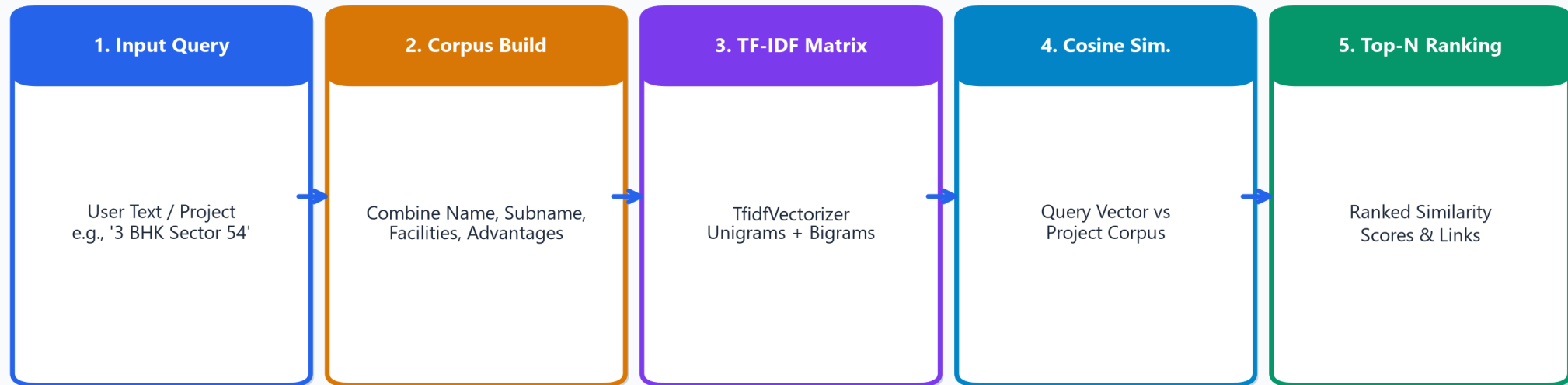
Price Prediction Model Workflow

Inference Pipeline Execution Steps

- **Step 1: Input JSON Payload:** FastAPI receives property features: property_type, sector, bedRoom, bathroom, balcony, agePossession, built_up_area, servant room, store room, furnishing_type, luxury_category, floor_category.
- **Step 2: Pydantic Validation:** Pydantic model verifies data types, default values, and converts JSON aliases to exact model feature names.
- **Step 3: ColumnTransformer Transformation:** Categorical variables pass through OneHotEncoder; numerical variables are standardized via StandardScaler.
- **Step 4: Random Forest Inference:** Fitted RandomForestRegressor (500 trees) calculates price estimate on log scale $\log_{10}(\text{price})$.
- **Step 5: Target Inverse Scaling:** TransformedTargetRegressor applies $\expm1(y_{\text{pred_log}})$ to return price on original Crore INR scale.
- **Step 6: API JSON Response:** FastAPI returns structured JSON: { 'predicted_price_crore': 2.45 }.

Content-Based Apartment Recommendation System

EstateIQ - Content-Based TF-IDF Recommendation Flow



Model Evaluation & Performance Metrics

R² Score (0.8211)

- **Metric:** R² Coefficient of Determination
- **Score:** 0.8211 (82.1%)
- **Interpretation:** The model explains 82.1% of the total variance in Gurgaon property prices.
- **Significance:** High predictive strength for complex real estate tabular data.

MAE (0.5304 Cr)

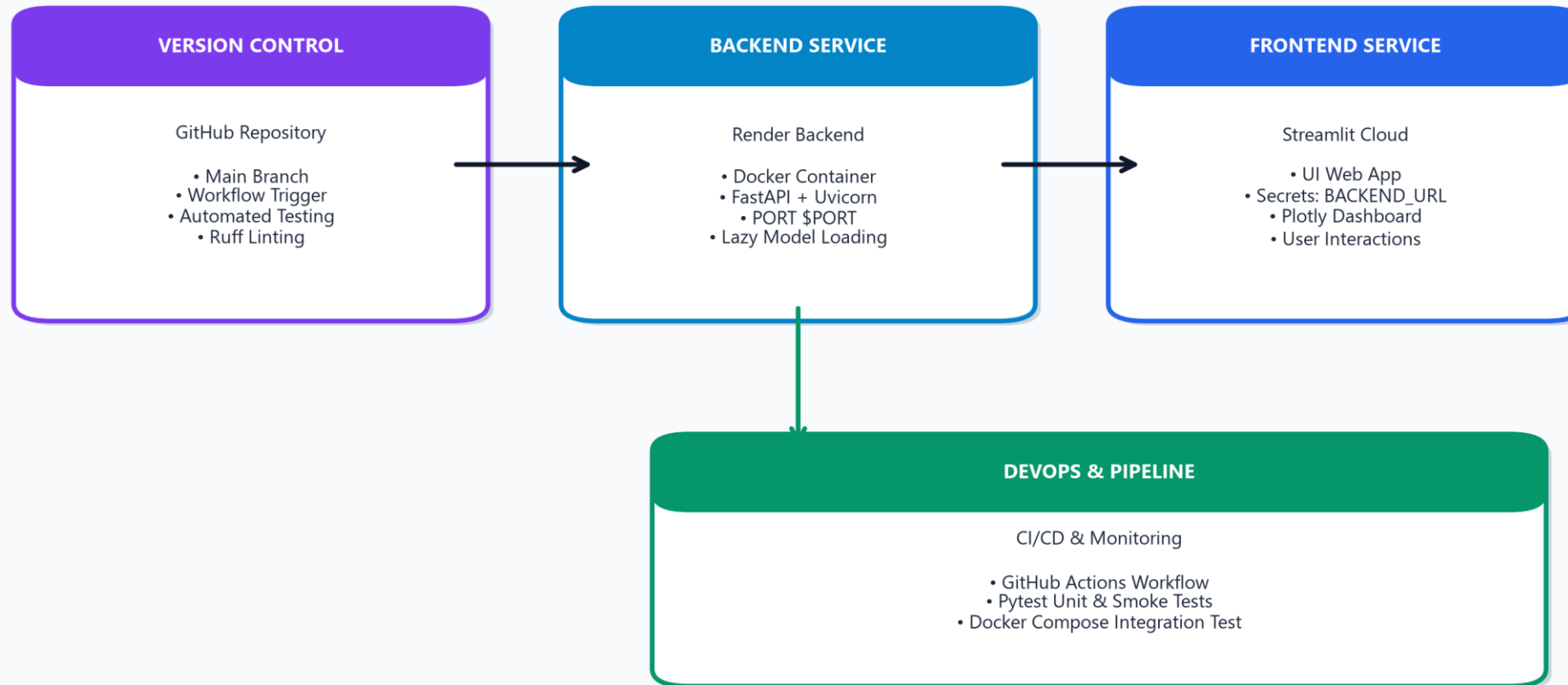
- **Metric:** Mean Absolute Error
- **Score:** 0.5304 Crore INR
- **Interpretation:** On average, model price predictions deviate by ~53 Lakhs INR from true prices.
- **Significance:** Highly usable for initial valuation and market screening.

RMSE (1.1727 Cr)

- **Metric:** Root Mean Squared Error
- **Score:** 1.1727 Crore INR
- **Interpretation:** Penalizes larger prediction errors occurring in high-end ultra-luxury properties.
- **Significance:** Reflects higher price volatility in top-tier luxury penthouses.

Deployment Architecture & Cloud Topology

EstateIQ - Production Deployment & CI/CD Architecture



Runtime System Flowchart

End-to-End Execution Trace

- **1. User Input in UI:** User selects city, sector, BHK, built-up area, furnishing, and luxury attributes in Streamlit web app.
- **2. HTTP POST Request:** Streamlit client sends JSON request payload over HTTP to FastAPI backend /predict endpoint.
- **3. Pydantic Verification:** FastAPI validates payload types, aliases, and defaults. Raises 400 Bad Request if schema is invalid.
- **4. ML Inference Execution:** PriceInferencePipeline transforms inputs via ColumnTransformer and passes data to RandomForest model.
- **5. Recommendation Trigger:** If user requests recommendations, TF-IDF engine computes cosine similarity over apartment text corpus.
- **6. Response & Dashboard:** Streamlit receives JSON output, renders predicted price card, price band, and Plotly analytics graphs.

User Interface Walkthrough & Page Navigation

Home Dashboard

- **Overview Cards:** Key platform metrics, property count, and project goals.
- **Locality Map:** Interactive map displaying sector-wise listings across Gurgaon.

Price Predictor Page

- **Input Form:** Clean dropdowns and numeric sliders for 12 property features.
- **Output Card:** Displays predicted valuation in Crore INR + estimated price range.

Apartment Recommender Page

- **Search Bar:** Free-text search by project name, locality, or facility keyword.
- **Property Cards:** Displays top matching apartment projects with similarity scores.

Market Analytics Page

- **Interactive Charts:** Price distribution histograms, area vs price scatter plots.
- **Heatmaps & Boxplots:** Sector median price rankings and luxury category breakdown.

Market Dashboard & Analytics Visuals

Interactive Analytics Features

- **Price Distribution Histogram:** Visualizes right-skewed pricing trends across residential properties.
- **Sector Price Heatmap:** Ranks top luxury sectors (Golf Course Rd, Sector 54, Sector 65 vs emerging sectors).
- **Area vs Price Scatter Plot:** Displays non-linear relationship between built-up area and property valuation.
- **Luxury Category Boxplots:** Compares price ranges across Low, Medium, and High amenity score tiers.

Business Intelligence Value

- **Investor Screening:** Allows investors to spot undervalued sectors and analyze price-per-sq.ft metrics.
- **Buyer Comparison:** Helps buyers benchmark seller listing quotes against sector median averages.
- **Broker Market Insight:** Provides agents with data-backed market visualisations for client consulting.

Future Scope & System Roadmap

Short-Term Enhancements

- **Deep Learning Models:** Experiment with TabNet & CatBoost for tabular pricing accuracy improvements.
- **LLM RAG Integration:** Add LangChain / Gemini API conversational assistant for natural language search.
- **Real-Time Listing Feeds:** Automate scrapers to pull live listing feeds and trigger retraining pipelines.

Long-Term Expansion

- **Multi-City Support:** Expand pricing pipeline to Noida, Delhi NCR, Bengaluru, and Mumbai.
- **Geospatial Distance Engine:** Calculate exact distance to metro stations, schools & tech parks.
- **Model Explainability UI:** Integrate SHAP waterfall plots into Streamlit for individual prediction breakdown.
- **Mobile Application:** Develop cross-platform Flutter mobile app for iOS & Android buyers.

Conclusion & Key Achievements

Project Summary & Milestones

- **End-to-End PropTech Solution:** Successfully converted messy real estate listing notebooks into a modular, production-ready ML platform.
- **High Predictive Performance:** Achieved strong validation accuracy with $R^2 = 0.8211$ and MAE = 0.5304 Crore INR.
- **Dual ML Capabilities:** Combined Random Forest price prediction with TF-IDF content-based apartment recommendation.
- **Production Architecture:** Built FastAPI REST services, Pydantic schema validation, and modular Streamlit UI.
- **DevOps & MLOps Excellence:** Implemented Docker containerization, GitHub Actions automated CI/CD, and metadata tracking.
- **Social & Industry Impact:** Delivered transparent, objective valuation tools for home buyers, brokers, and investors.